13 Alex wants to find out how many ducks there are in a park.

One day he puts a tag on each of 30 of the ducks.
The next day he catches 40 ducks.
8 of these ducks have tags on them.

(i) Work out an estimate for the number of ducks in the park.

Assume the tagged proportion is the same in the sample and in the park. Tags means tagged ducks; All means all ducks.

1 mark
Compare proportions

	Tags	All	
	8	40	
× 3.75			× 3.75
	30	150	

30 ÷ 8 = 3.75

40 × 3.75 = 150

1 mark
Scale sample to park

1 mark
150 ducks

Alex assumed that none of the tags fell off during the night.

(ii) If Alex's assumption is wrong, explain how this could affect your answer to part (i).

Lost tags would make the tagged fraction too small, so 150 would overestimate the number of ducks.

1 mark
Estimate too high

(Total for Question 13 is 4 marks)